Research Innovation and Technical Writing CSE414 (Section: 65 C/E)

https://www.linkedin.com/company/quantum-robotics-automation-research-group/?viewAsMember=true

Lecture-7

AI & ML

Instructor

Dr. M. Akhtaruzzaman
Associate Professor and Head, RME, DIU
akhtaruzzaman.cse@diu.edu.bd; akhter900@gmail.com; akhter900@qrarg.com

June 21-24, 2026, DIU, Dhaka.

رب زدني علما وَارْزُقْنِيْ فهما.

My Lord! Advance me in Knowledge and True Understanding.

Content

- Short intro
- Industrial Revolution
- Era of AI & IoT
- Bangladesh perspectives
- AI in IR4.0
- Robotics
- GAI, XAI, GAN
- Reinforcement Learning
- Hands-on with AKI (ML Tool)
- Future (IR5.0)!!!

INTRODUCTION TO INSTRUCTORS

QRARG
Quantum Robotics & Automation Research Group

M. Akhtaruzzaman, PhD

Doctor of Philosophy in Engineering (Mechatronics/Robotics) , IIUM, Malaysia

MSc in Mechatronics Engineering, IIUM, Malaysia

BSc in CSE, IIUC, Bangladesh

Research Interests

- Robotics
- Mechatronics Systems
- Artificial Intelligence
- Machine Learning

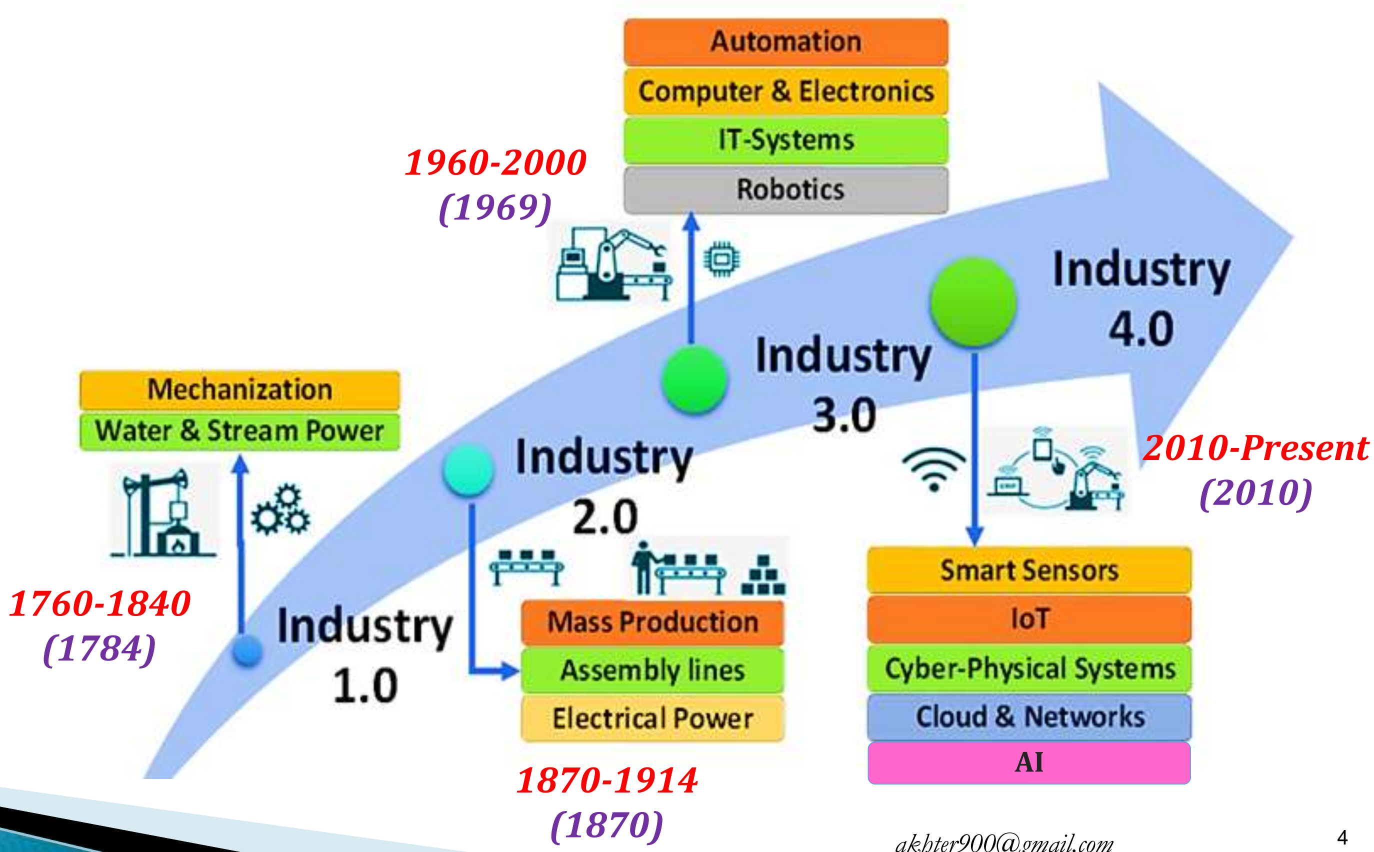
Publication & Citation Summary

- Total publications: 85+
- Citation: 1080, H-index: 16, i10-index: 22

akhter900@gmail.com

INDUSTRIAL REVOLUTION

Industrial Revolutions are time periods with **rapid advancements** in **production technology** which greatly **transform the industrial procedures** and **economic factors** of the **world**.



akhter900@gmail.com

IR4.0

Industry 4.0 is the new phase in industrial revolution that introduce AI based networking of machines and processes for industry with the help of information and communication technology.

Rock-Paper-Scissor (interactive human-robot zero-sum game)

https://www.youtube.com/watch?v=fuv0FTcmA5I&t=24s

https://www.linkedin.com/posts/quantum-robotics-automation-research-group_httpslnkdingr6qfhzf-the-loving-child-activity-7311342686597193728-qvwl?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAujliUBGkTovaDvOdsmXXbEViUmSkpiOSU

Era of IR4.0

The era of AI & IoT

Simulation
Big data analytics
CIoT IoT Robotics
IIoT AI-IoT System integration
Cloud computing UAV UGV
Additive manufacturing (3D printing)
Augmented reality Cyber security
Virtual reality Artificial Intelligence
Digital twin & Metaverse Generative AI
ML DL XAI GAN
Reinforcement Learning

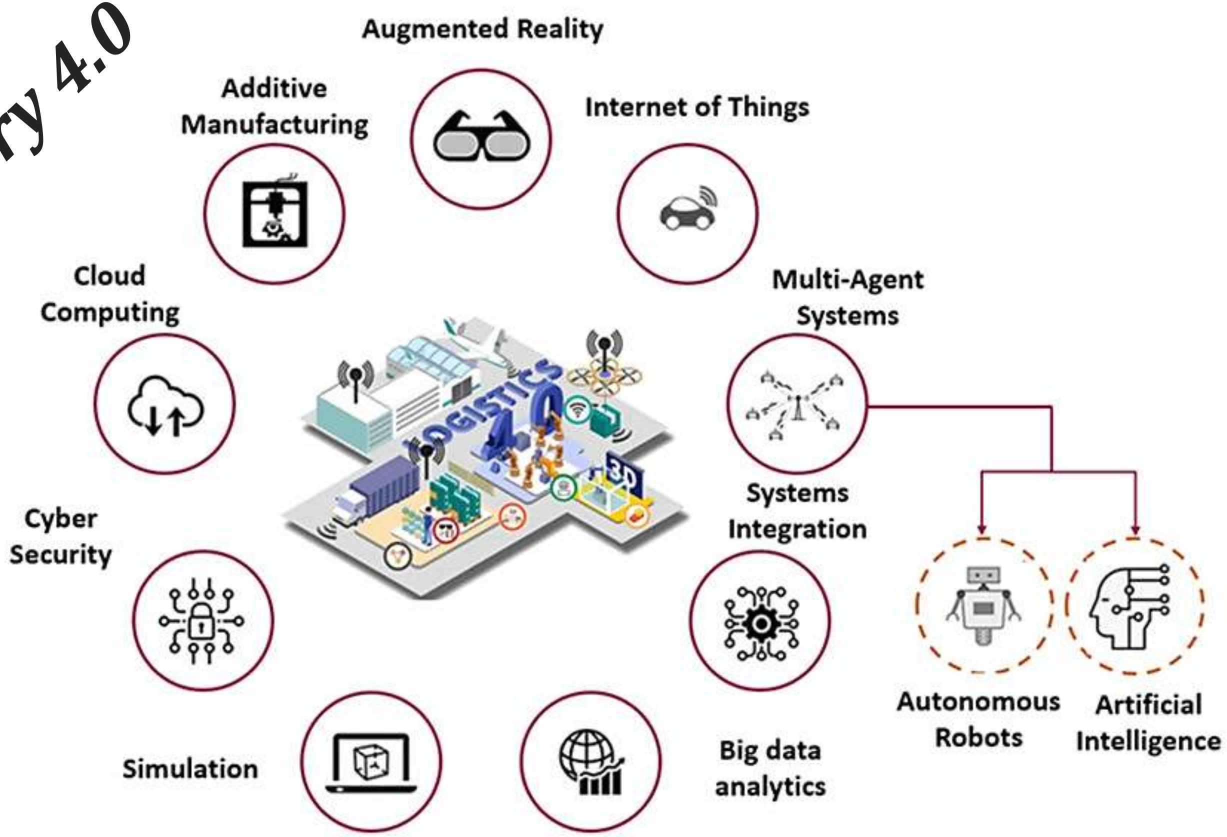




akhter900@gmail.com

Era of IR4.0…

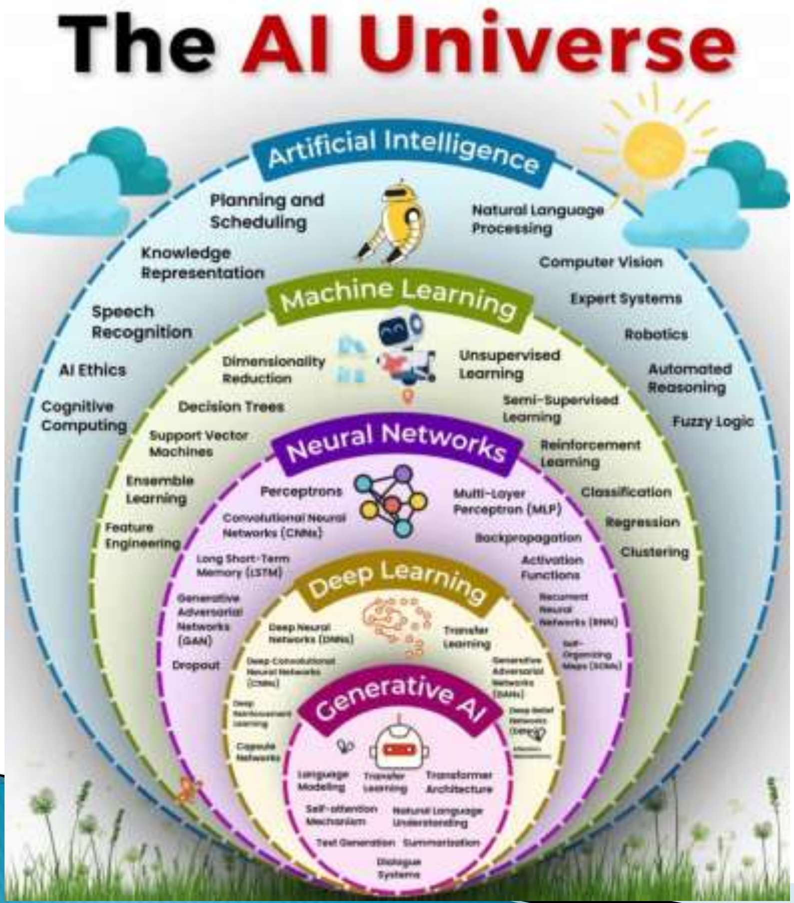
11 pillars of Industry 4.0



akhter900@gmail.com

AI & IR4.0

AI & IR4.0



Industry 4.0

is all about

Digitization & Automatic manufacturing

through

interconnected technologies

that enable

smarter, more efficient, and flexible production processes.

It has the **potential** to **transform entire industries** by **increasing innovation**, **reducing costs**, and **improving sustainability**.

Values created from IR4.0

automating standard tasks** and **reducing manual interventions

Process efficiency

automating job function** and then undertaking **job substitutions

Cost reduction

***Increase revenue** through **hyper-personalized marketing** for target customers*

Growth

AI/IR4.0

Government citizen services

***Increase accuracy** of various **federal and local programs** and create **easier access for at-risk populations**.*

Accelerating Innovation

*Increase the **pace of new product or new service development** and speedier **go-to-market.***

New discovery and Insights

***Uncover new ideas**, **insights**, and **questions** and generally **unleash creativity**.*

akhter900@gmail.com

Challenges of IR4.0

- **Healthcare sector:**
  - **prioritizing the confidentiality of patient** data is supreme.
  - **Precision in medical consultations** to avoid misdiagnosis or dissemination of misinformation.
- **Financial domain** (**detecting fraud** during financial transactions)**:**
  - The challenge is to enable generative AI to **expertly recognize and counter evolving and intricate fraud patterns**.
- **Customer service arena:**
  - **AI-generated responses are timely and contextually appropriate** ensuring that for real-time interactions.

Challenges of IR4.0...

- **Education:**
  - AI is leveraged to customize learning experiences for students. However, the challenge lies in **crafting content related to individual student needs** and learning styles while upholding **ethical guidelines and promoting positive educational values**.
- **Verifying the accuracy of AI-generated quality control** reports is crucial to prevent the shipment of defective products.
- **Entertainment sector:**
  - Balance between AI-generated creativity in **music, art, literature and human artistic expression**.
  - Averting **copyright infringement and plagiarism** in AI-generated content is a significant concern.

Bangladesh Perspective

- Industry 4.0 is being introduced in several **key sectors of Bangladesh** like the **RMG**, **textile**, **agro-processing**, **leather**, **footwear**, and **tourism industry**.
- **Light engineering**, **furniture manufacturing**, **shipbuilding and breaking**, **tourism**, and **hospitality sectors** of Bangladesh have made significant investments in **robotics, sensors, software, automated machinery, IoT, and big data**.

Bangladesh Perspective…

- Challenges
  - **Lack of motivation, awareness, infrastructure, skills, investment, and technology**
  - Bangladesh is still **largely dependent on *manual labor*** in regard to the manufacturing industries.
  - **Public and private enterprises** need to **take active steps to improve** the policies, investments, infrastructure, education, and training so that Industry 4.0 can be adopted into the manufacturing and service industries of the country.

Bangladesh Perspective…

- Barriers
  - **Lack of Motivation**
  - **Resistance to Change**
  - **Lack of technical expertise**
  - **High initial investment costs**
  - **Weak infrastructure**, **availability of cheap labor**, **costly installation of technologies**, **lack of policy and supports** and most importantly **lack of skills**.

The **benefits of implementing Industry 4.0** in Bangladesh need to be described specially in **garments sector** which is one of the most valuable exporting sectors of Bangladesh

Bangladesh Perspective…

❑ Solutions:

- **Invest** in education and **upskilling** in AI and automation.
- **Government subsidies and incentives** for adopting IR4.0.
- **Public-private partnerships** for sustainable tech deployment.
- Stop the **Brain-drain.**
- **Being strong in Ethical Mental State**

Example Steps

- Garment Industry:
  - **AI-powered visual inspection** in factories to detect defects in fabrics, reducing waste and ensuring quality.
  - Generative AI for **customized clothing design**.

- Banking Sector:
  - **ChatGPT-like AI tools** for customer support in local banks, streamlining queries in Bangla.
  - **Fraud detection** through anomaly analysis.

Example Steps…

- **Healthcare:**
  - **AI-generated 3D models** for custom medical devices.
  - **Automating patient records and diagnosis** with natural language processing (NLP).

- **Agriculture:**
  - **AI-driven weather prediction** and **crop recommendation systems** for farmers.
  - Generative AI for **creating training videos** in local dialects.

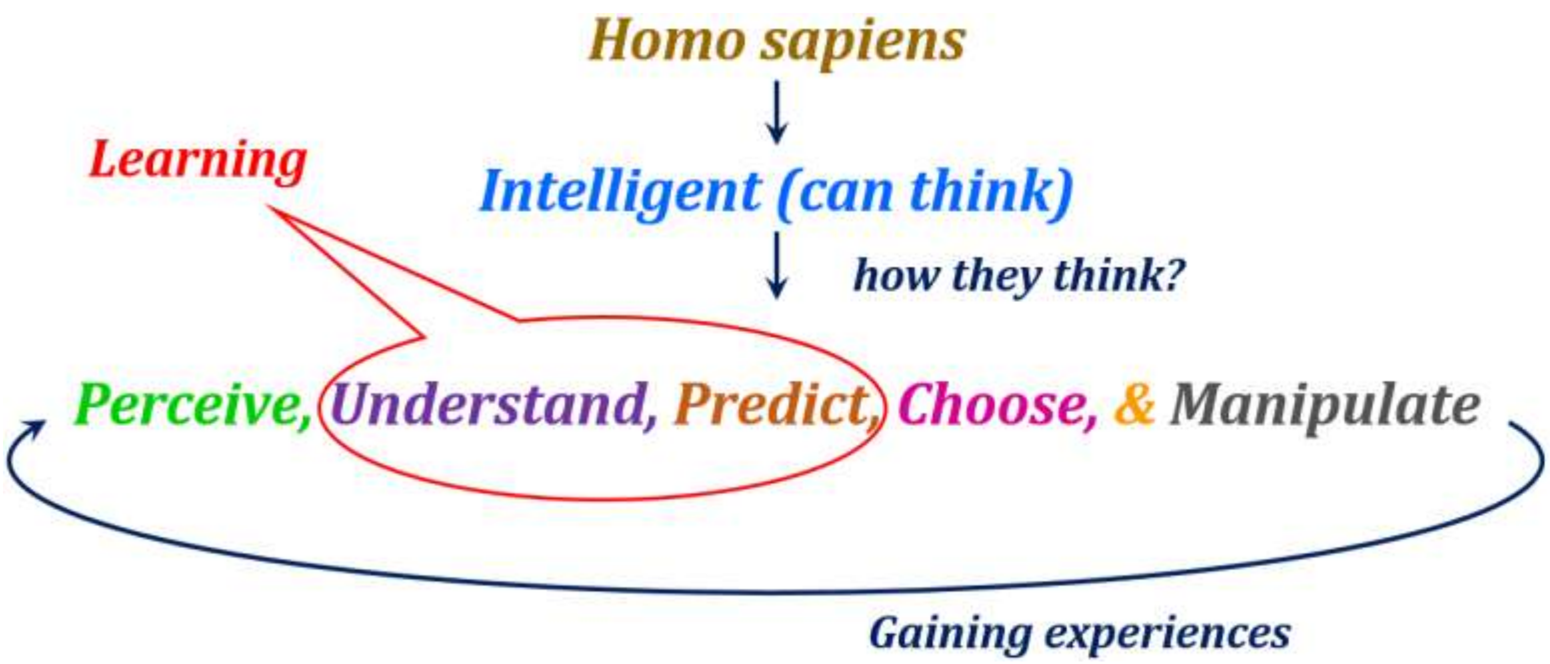
Benefits for Bangladesh

- **Economic Impact:**
  - Increased productivity, reduced costs, and global competitiveness.
- **Job Creation:**
  - New roles in AI development, maintenance, and Industry 4.0 solutions.
- **Sustainability:**
  - Smart solutions to manage energy and waste effectively.
- **Accessibility:**
  - AI solutions customized for local languages and needs.

What is AI ?
What is ML ?

Why AI is important ?

Intelligence & Learning



7 MAJOR TYPES OF INTELLIGENCE

- Linguistic intelligence
- Musical intelligence
- Logical-mathematical intelligence
- Spatial intelligence
- Bodily-Kinesthetic intelligence
- Intra-personal intelligence
- Interpersonal intelligence

What is AI?

A **smart Model/Agent** which
can perceive,
acquire knowledge,
able to do reasoning,
able to judge rationally and take decisions,
and
able to act accordingly to its environment.

AI & ML

- Artificial Intelligence (AI):
  It is a process of creating intelligence for decision making by extracting knowledge from history, where learning is done by machine instead of human.

- Machine Learning (ML):
  Mathematical algorithms that helps learning from history, extract knowledge then either classifies or quantifies

Importance of AI & ML

AI is the latest technology that augments decision making process.

AI puts decision making at a NEW level.

Application of AI & ML are in all sectors of the economy that affects life

Robotics

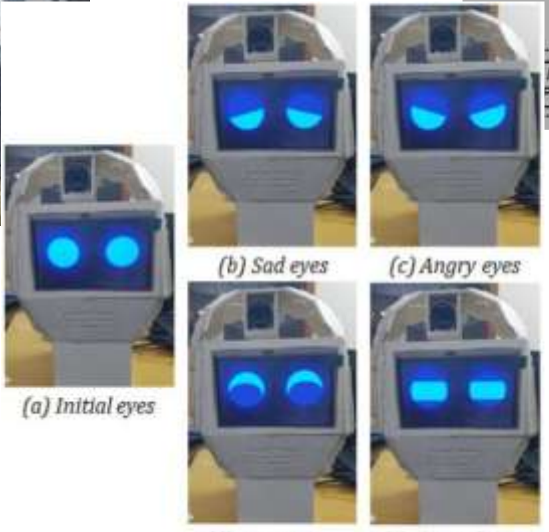
Robotics

A ROBOT is an AI integrated machine that can perform tasks with little or no human intervention.

ROBOTICS is the field of study that focuses on the design,

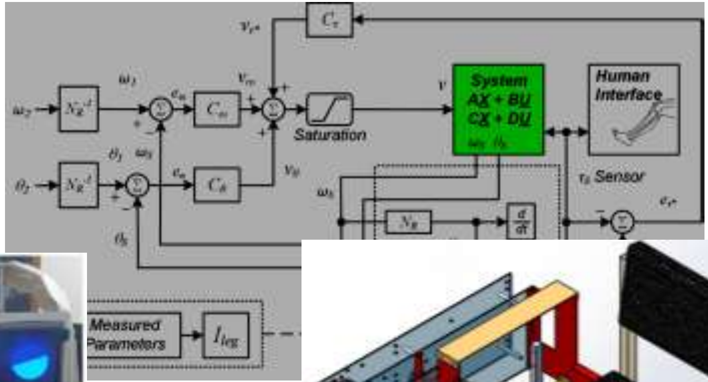
Robotics in IR4.0

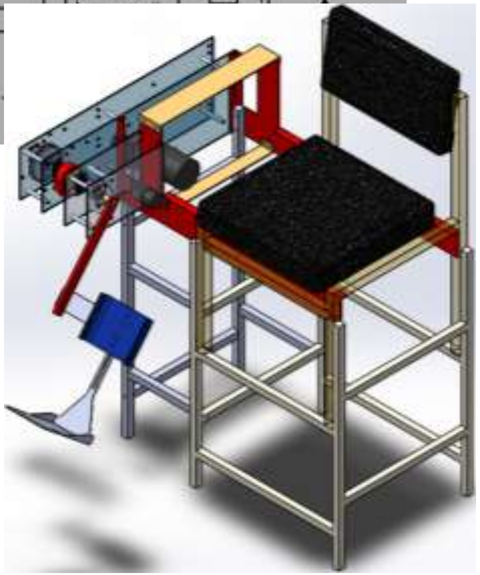
Robotics in IR4.0…



(a) Initial eyes (b) Sad eyes (c) Angry eyes (d) Happy eyes (e) Normal eyes

https://mijst.mist.ac.bd/mijst/index.php/mijst/article/view/214





https://mijst.mist.ac.bd/mijst/index.php/mijst/article/view/440

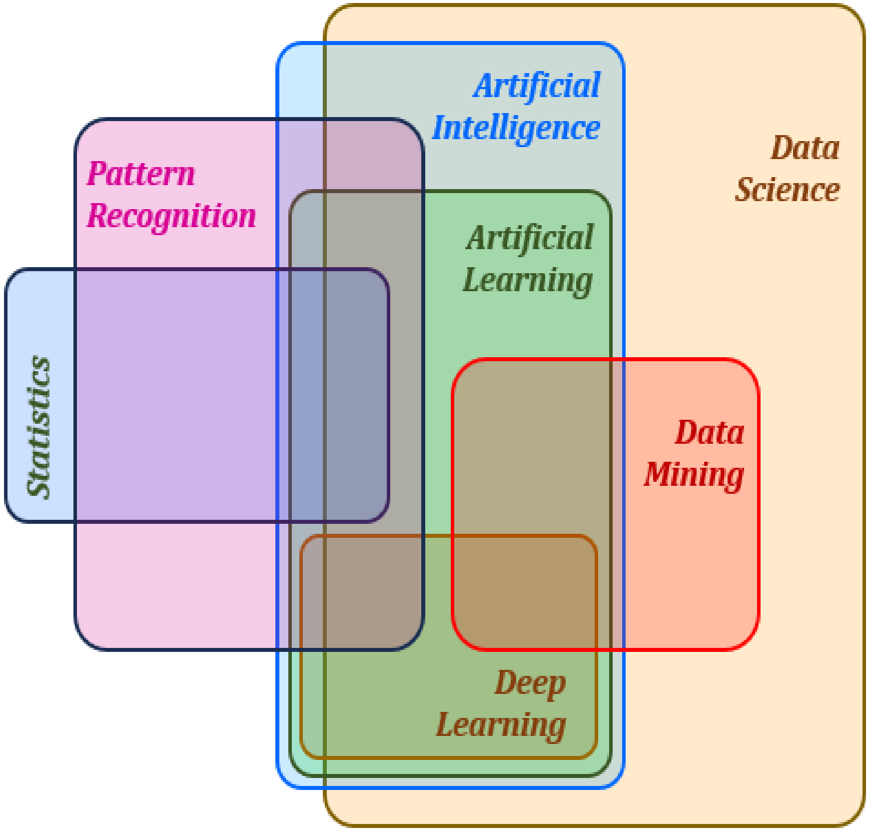
akhter900@gmail.com

Machine Learning (ML)

Machine Learning (ML)

Learning capabilities of an Artificial System

for quantification or classification



Example Scenario 1

- Mr. Z is a Successful **businessman.**
- Mr. Z wants to **expand his business** portfolio, in used car sales.
- He found out, **used car market** will grow over the next years in Bangladesh.
- Business **plan** is simple:
  buy cars in good condition from private sellers and **sell** them with a profit.
- Mr. Z starts the business by:
  - Taking a **loan** from bank.
  - Buy cars and **sell** them for a profit.
- Unfortunately, after one year, his car business is suffering **heavy losses**.

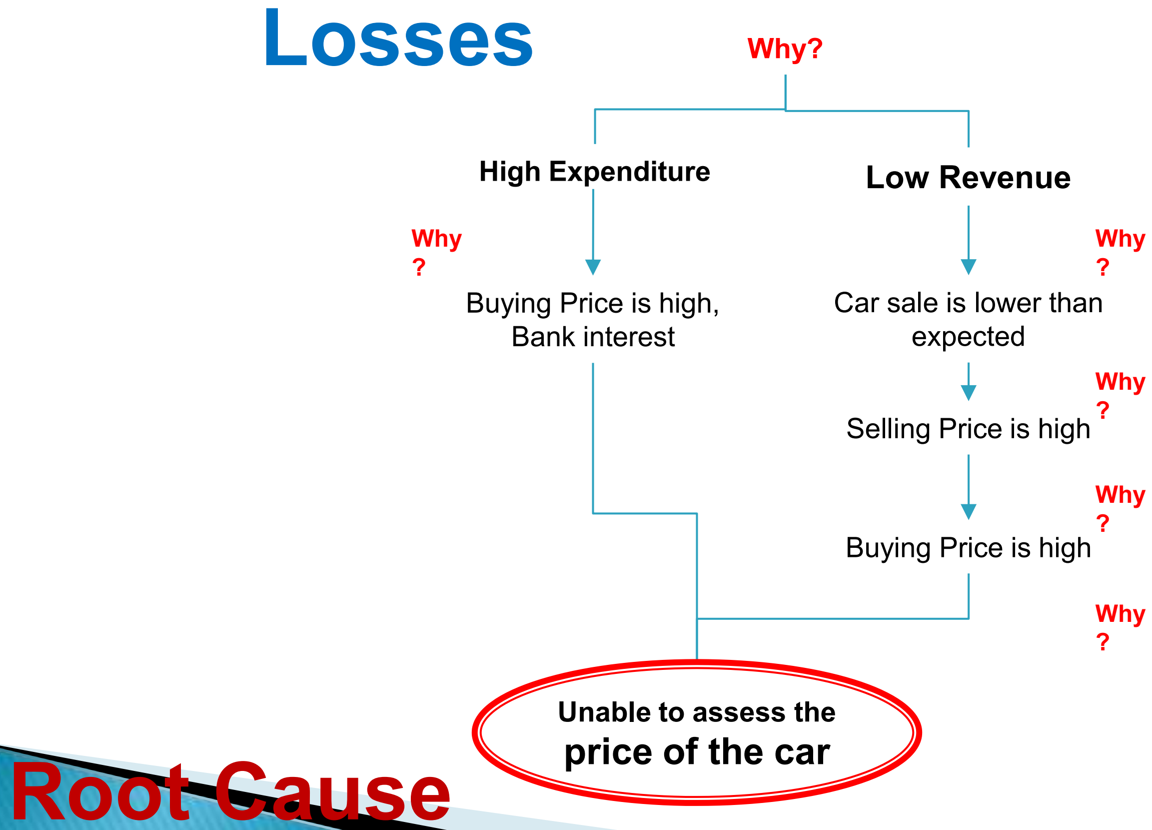
Example Scenario-(Problem Identification)

- After one year, Mr. Z's car business is suffering **heavy losses**.

Problem → **Heavy Losses**

Why?

Example Scenario–(Identifying Root Cause)



Example Scenario–(Identifying Solutions)

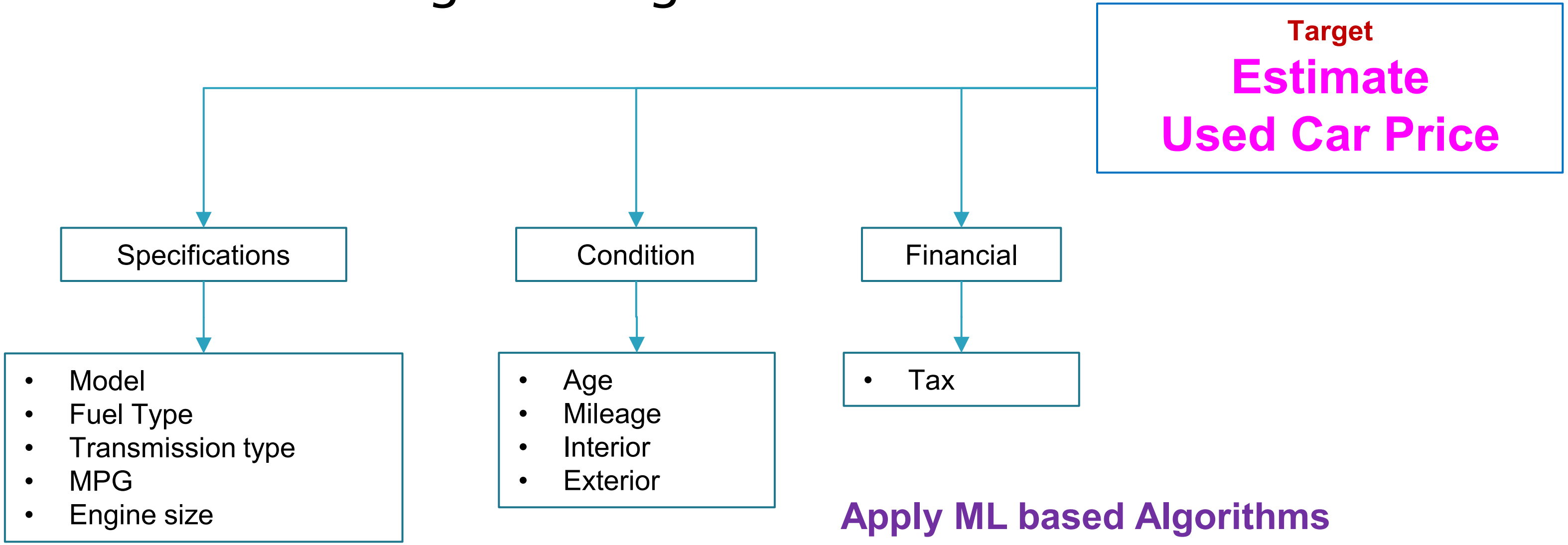
- Solution is, (what **Mr. Z needs to know**):
  - **Approximate price of a car** from a private seller
  - **Decide if he can make a profit** on it.

How to do those?

- **Hiring a Business Analyst?**
- **Hiring a Data Scientist?**
- **Learn and conduct Statistical Analysis?**
- **Hiring MIST ML Engineer for an Intelligent solution?**

Example Scenario–(ML based Solutions)

- Identify data points (features) required to address Target
- Do Critical thinking on Target



Apply ML based Algorithms
(may be using your own AI Tools)
and
Provide the best solutions in decision making

integration...)

$$\textit{Questions}\left\{\begin{matrix}\textit{persistent cough}\\\textit{persistent fever}\\\textit{night sweats}\\\textit{weight loss}\\\textit{lump in neck}\\\textit{TB History}\end{matrix}\right\}\rightarrow\rightarrow\rightarrow \textit{Screening Result}\ (Y/N)$$

1. **General process (Manual)**
2. **Web/App process (Digital)**
3. **Web/App Chatbot (Digital)**

4

$$\textit{Questions}\ (\textit{Chatbot ML})\left\{\quad\right\}\rightarrow\rightarrow\rightarrow \textit{Screening Result}\ (Y/N)$$

integration...)

How to implement?

- Hiring a Business Analyst?
- Hiring a Data Scientist?
- Learn and conduct Statistical Analysis?
- Hiring MIST ML Engineers for an Intelligent solution?

Scenario-2 (Example Dataset)

- **Chatbot (NLP + ML):** AI enabled system with Natural Language Processing (NLP) and Machine Learning (ML)

SMOTE" stands for "Synthetic Minority Over-sampling Technique

Features/Independent variable (X)

Target/Dependent variable (Y)

Having 2 classes, Yes & No

Sample Dataset

Train Dataset (80%)

Test Dataset (20%)

patient	cough	fever	night_sweats	weight_loss	tb_history	lump	result
p43	No	Yes	No	No	Yes	No	Yes
p44	No	Yes	No	No	No	Yes	Yes
p45	No	No	Yes	Yes	Yes	No	Yes
p46	No	No	Yes	Yes	No	Yes	Yes
p47	No	No	Yes	No	Yes	No	Yes
p48	No	No	Yes	No	No	Yes	Yes
p49	No	No	No	Yes	Yes	Yes	Yes
p50	No	No	No	Yes	Yes	No	Yes
p51	No	No	No	Yes	No	Yes	Yes
p52	No	No	No	No	Yes	Yes	Yes
p53	No	No	No	No	No	Yes	Yes
p54	No	No	No	No	No	No	No
p55	No	No	Yes	No	No	No	No
p56	No	Yes	No	No	No	No	No
p57	No	No	No	No	Yes	No	No
p58	No	No	No	Yes	No	No	No
p59	No	Yes	No	No	No	No	No
p60	No	No	Yes	No	No	No	No

Record of Patient p51

Splitting and Cross-validation

Class wise splitting

80%-20%

70%-30%

ϕ based splitting

5-Fold validation

K-Fold validation

SMOTE (Synthetic Minority Over-sampling Technique)

Class wise splitting (80%-20%, 70%-30%, ϕ based splitting)

akhter900@gmail.com

Evaluating Model Performances

- Model Evaluation for Classification problem
  - $$Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ Predections}$$
  - $$Precision = \frac{True\ Positive}{Total\ PREDICTED\ positive}$$
  - $$Sensitivity = \frac{True\ Positive}{Total\ ACTUAL\ positive}$$
  - $$F1\ score = 2 \times \frac{Precession \times Sensitivity}{Precession + Sensitivity}$$
  - ROC-AUC (receiver operating characteristic)
  - Log Loss (Cross-entropy loss)

Confusion matrix

	N=165	Predicted (based on Test data): Yes/High/T/1/+ ($TP + FP = 105$)	Predicted (based on Test data): No/Low/F/0/- ($FN + TN = 60$)
Actual or true values of Test cases	Yes/High/T/1/+ ($TP + FN = 110$)	100 *TP*	10 *FN*
	No/Low/F/0/- ($FP + TN = 55$)	5 *FP*	50 *TN*

akhter900@gmail.com

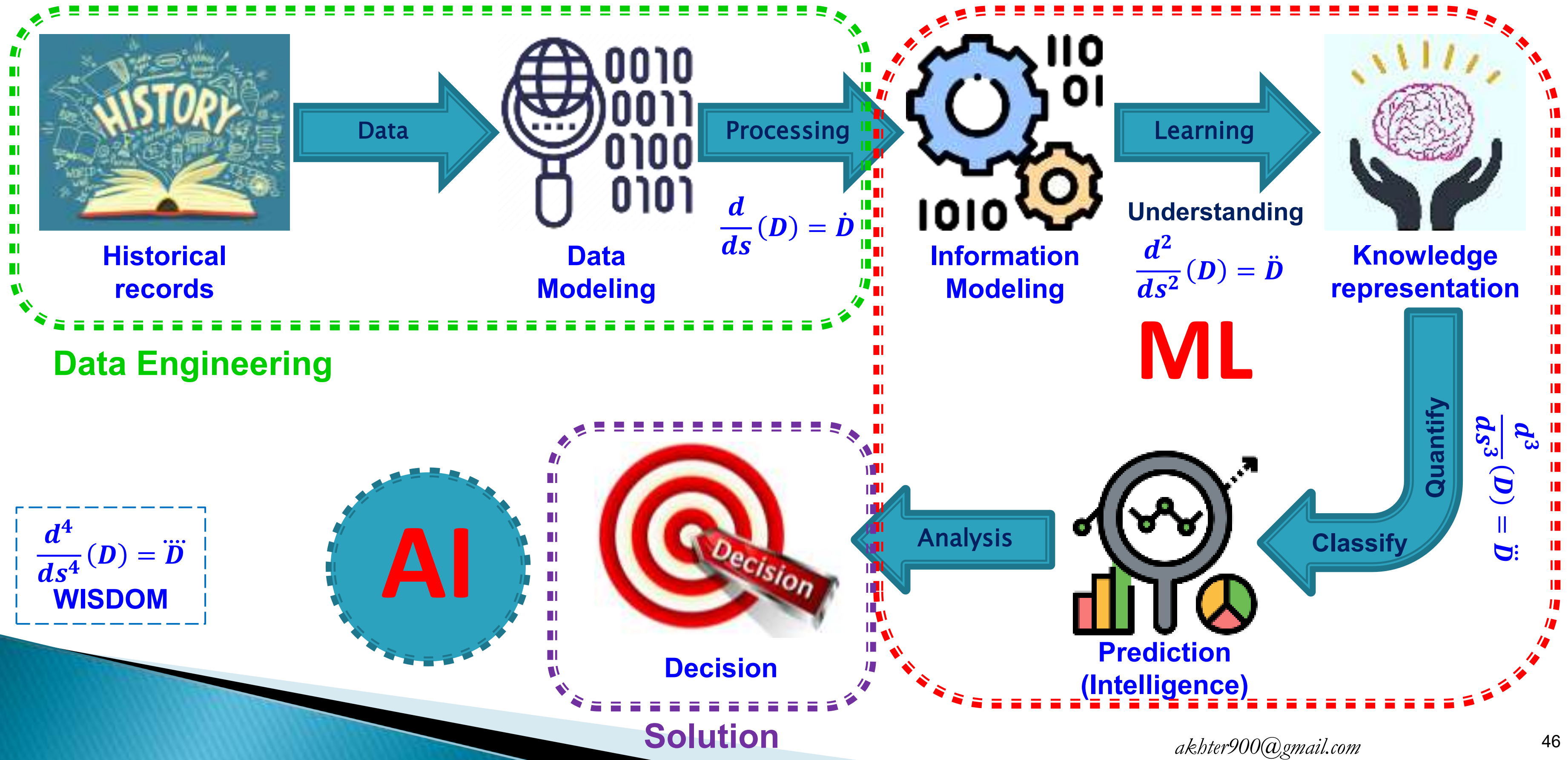
Evaluating Model Performances

- Model Evaluation for Regression problem
  - $MAE = \left(\frac{1}{n}\right) \Sigma_{i=1}^{n} |Actual - Predict|$
  - $MSE = \left(\frac{1}{n}\right) \Sigma_{i=1}^{D} (Actual - Predict)^2$
  - $RMSE = \sqrt{\left(\frac{1}{n}\right) \Sigma_{i=1}^{D} (Actual - Predict)^2}$
  - $R^2 = 1 - \left(\frac{\Sigma(Actual - Predict)^2}{\Sigma(Actual - Actual\ Mean)^2}\right)$

Minimize

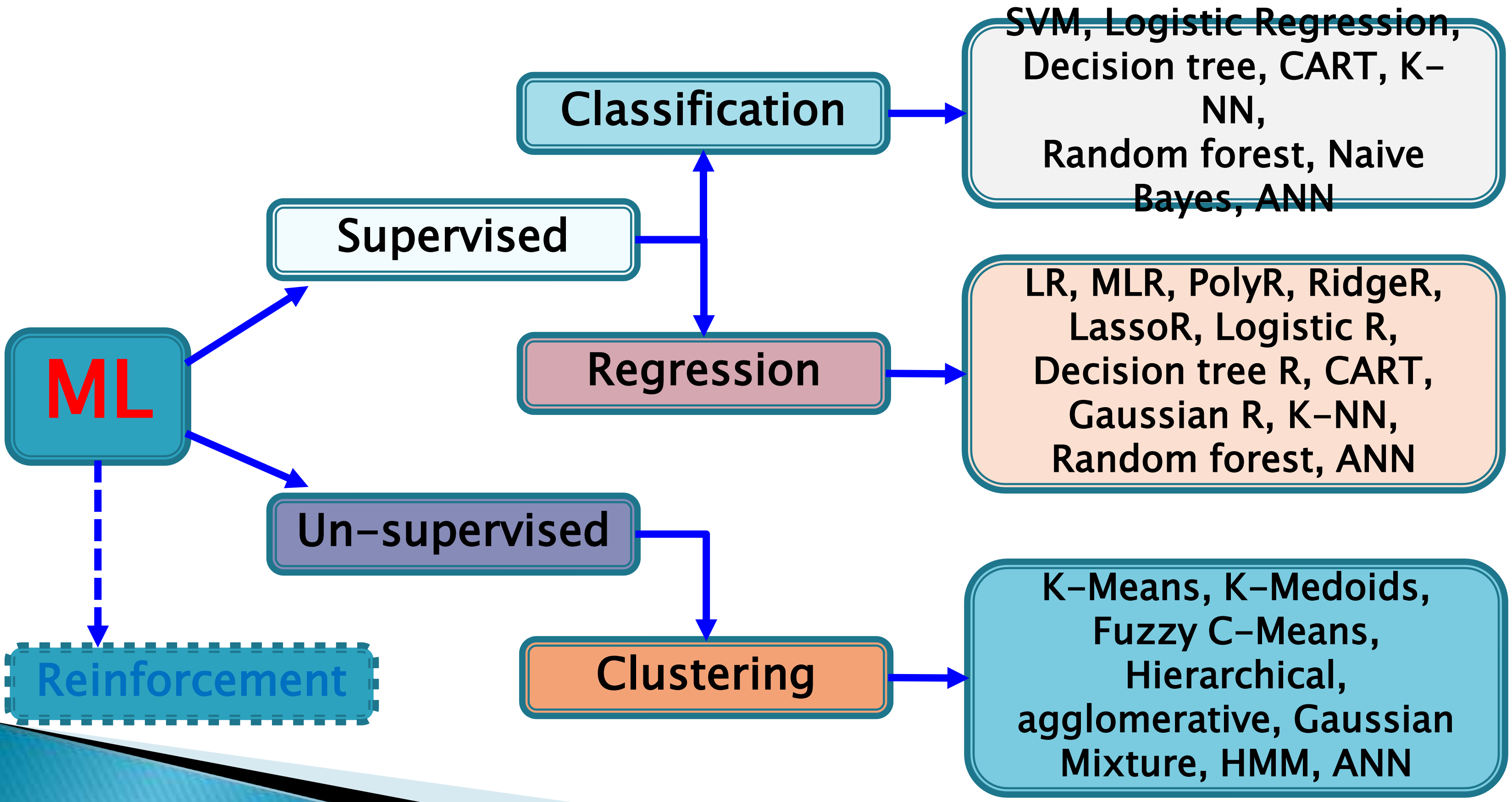
Maximize

The Big Picture of AI & ML

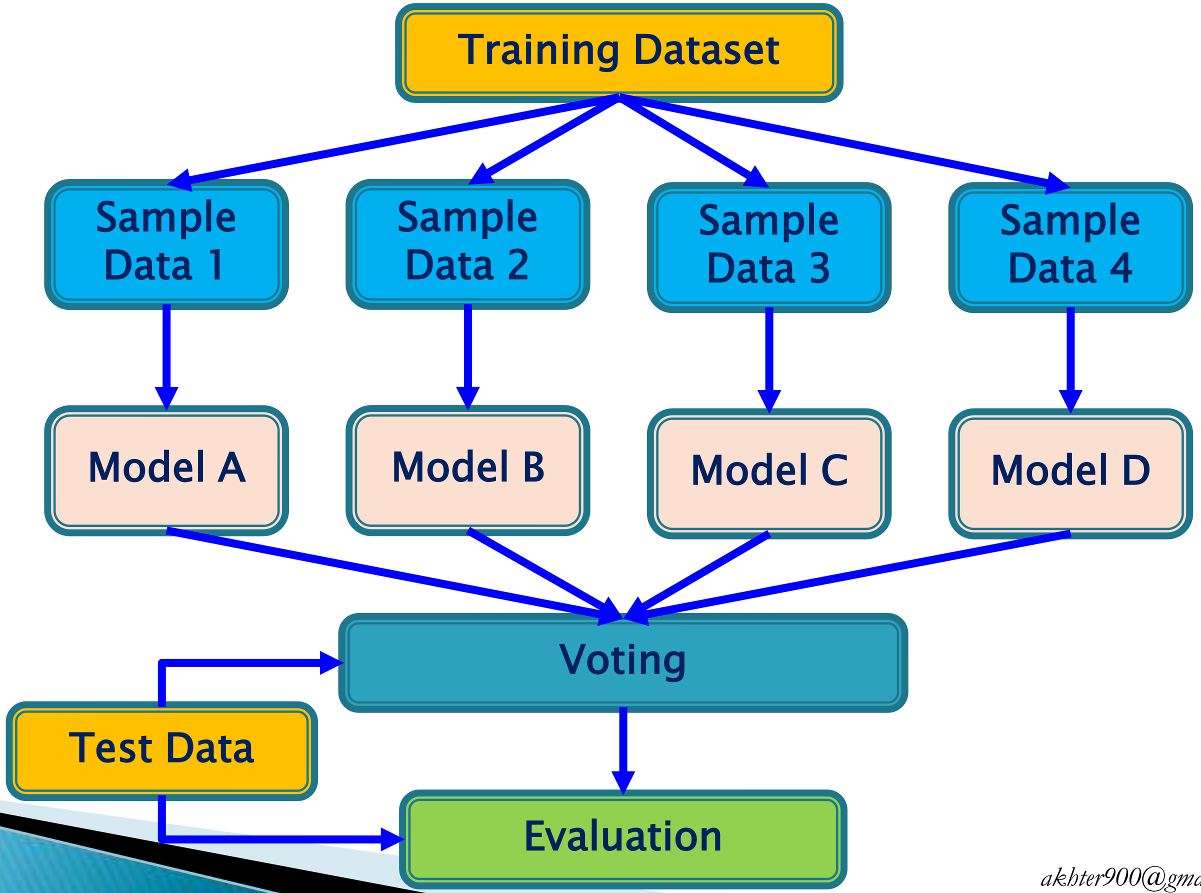


akhter900@gmail.com

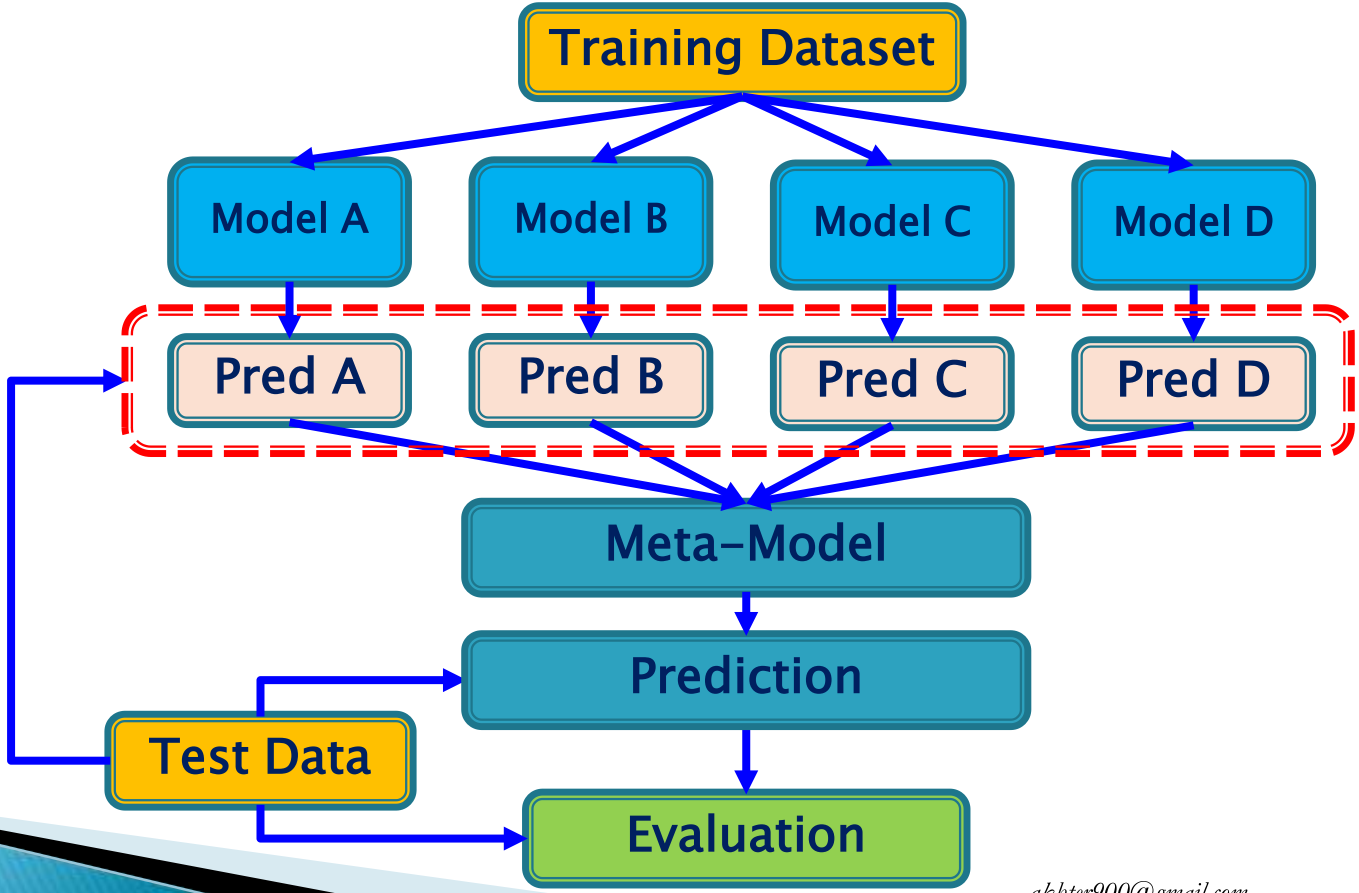
Categories of ML



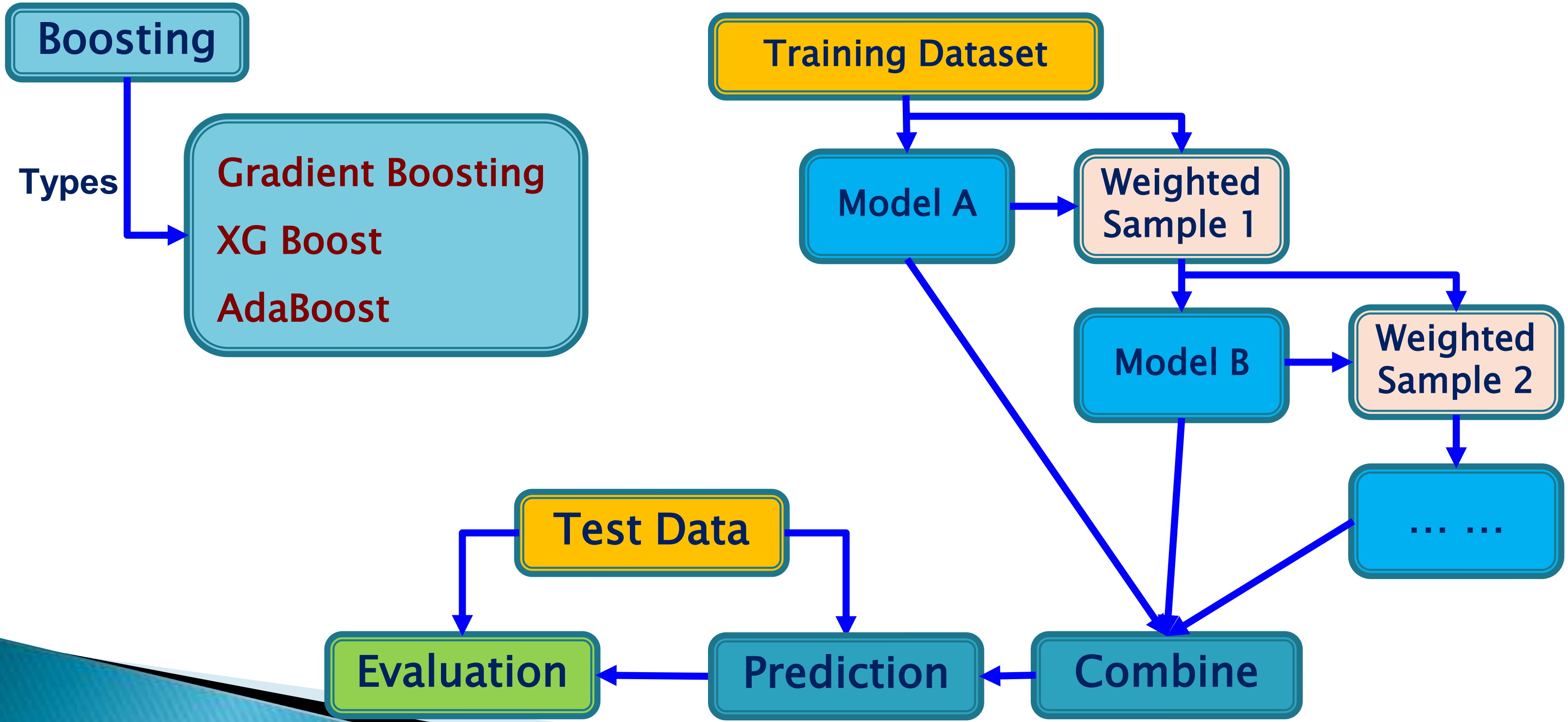
Ensemble Learning (Bagging)



Ensemble Learning (Stacking)



Ensemble Learning (Boosting)

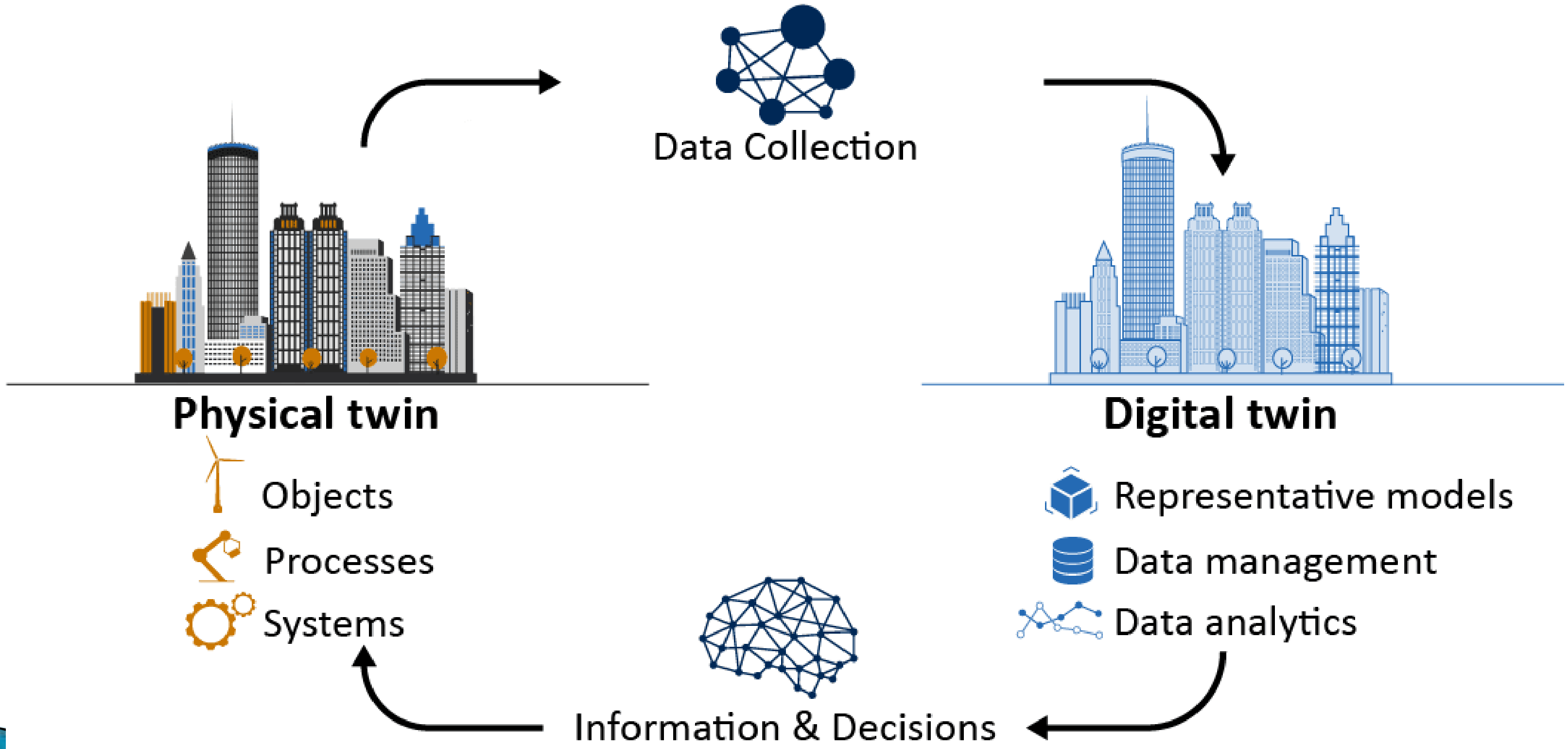


akhter900@gmail.com

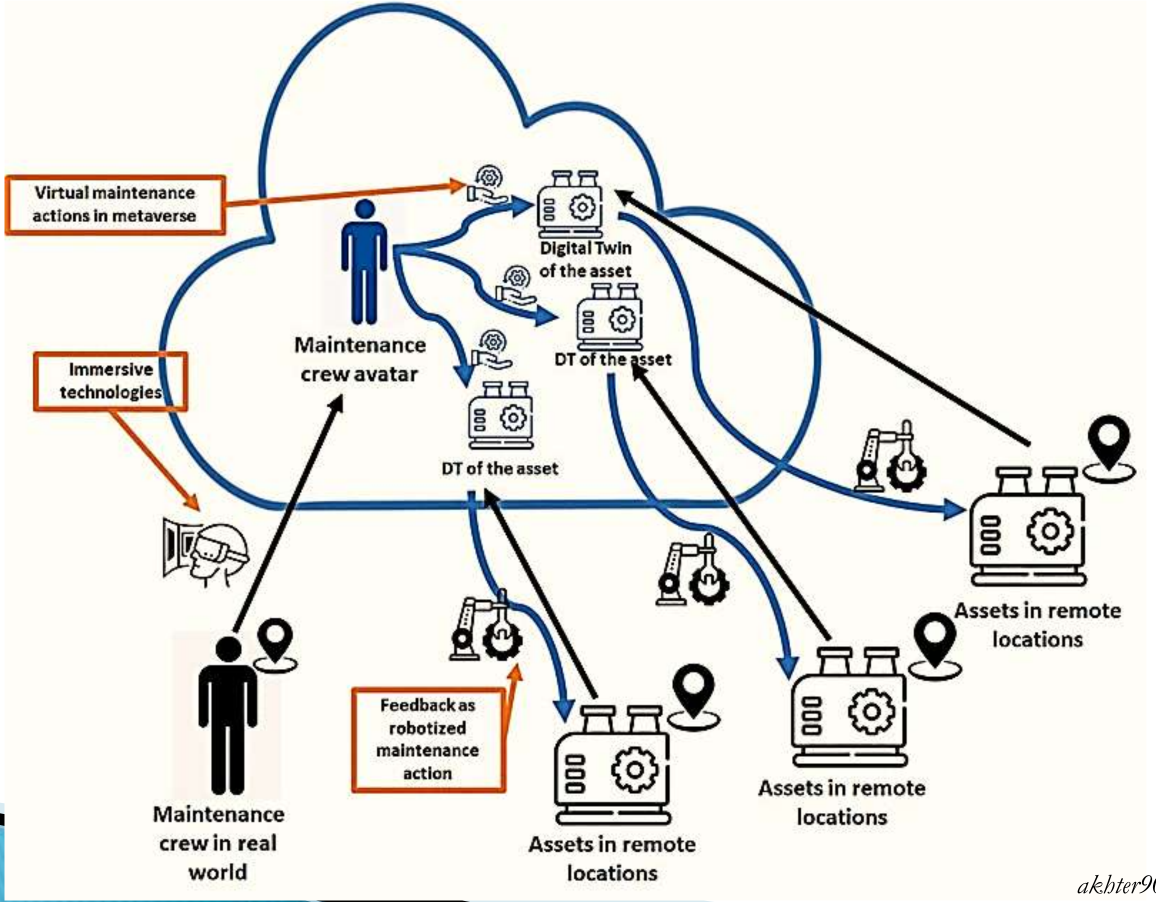
Latest Technology Advanced to
Digital Twin
&
Metaverse *(Digital Twin + Avatar)*
where AI & ML serves in the background

akhter900@gmail.com

Digital Twin



Metaverse

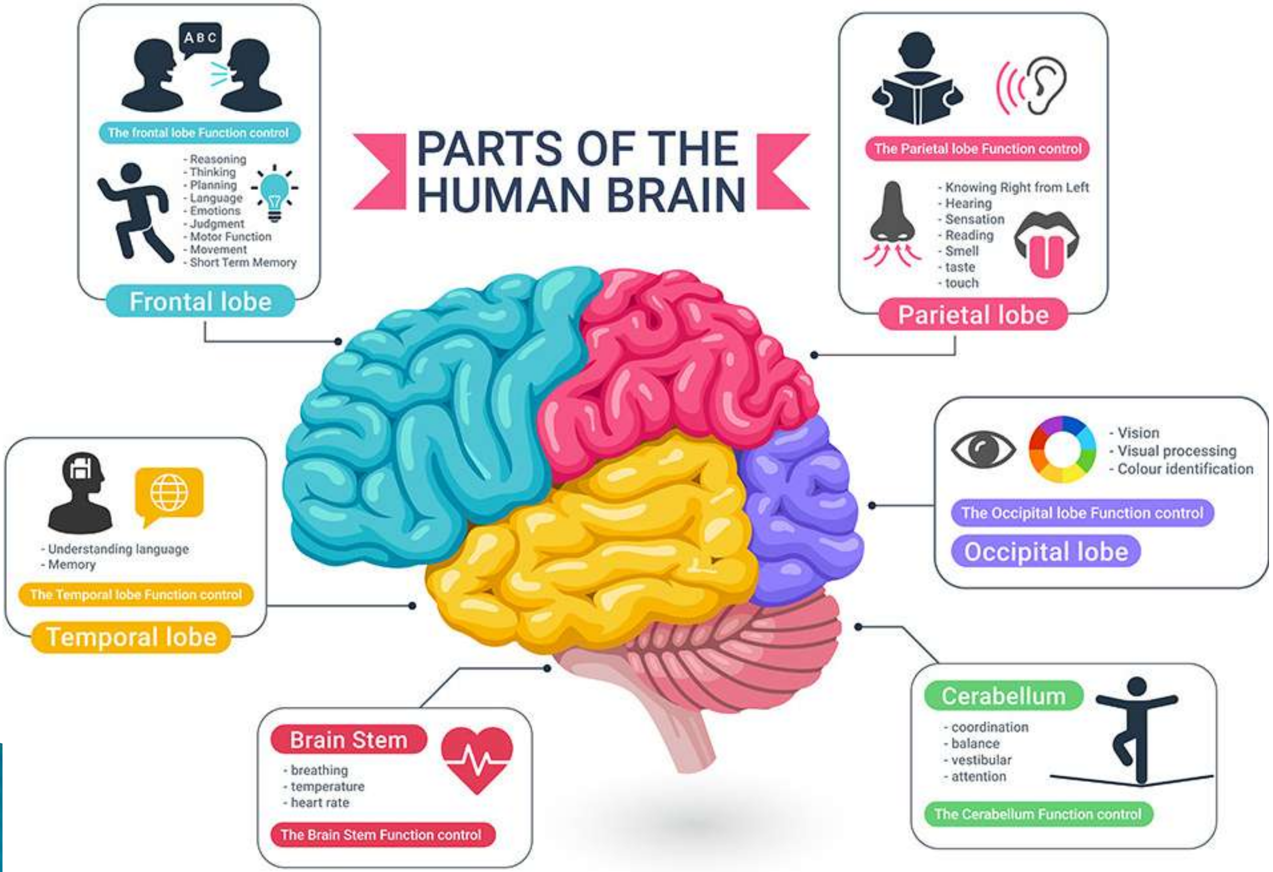


Virtual maintenance actions in metaverse
Digital Twin of the asset
Maintenance crew avatar
DT of the asset
Immersive technologies
DT of the asset
Assets in remote locations
Feedback as robotized maintenance action
Maintenance crew in real world
Assets in remote locations
Assets in remote locations

Generative AI (GAI)

https://www.youtube.com/watch?v=ctWfv4WUp2I

Human brain



Human brain

The human brain contains many neurons and synapses, estimated to be

10^{11} neurons

&

10^{14} synapses

The Marvel of the human brain: Unravelling neural networks

Dr M Akhtaruzzaman

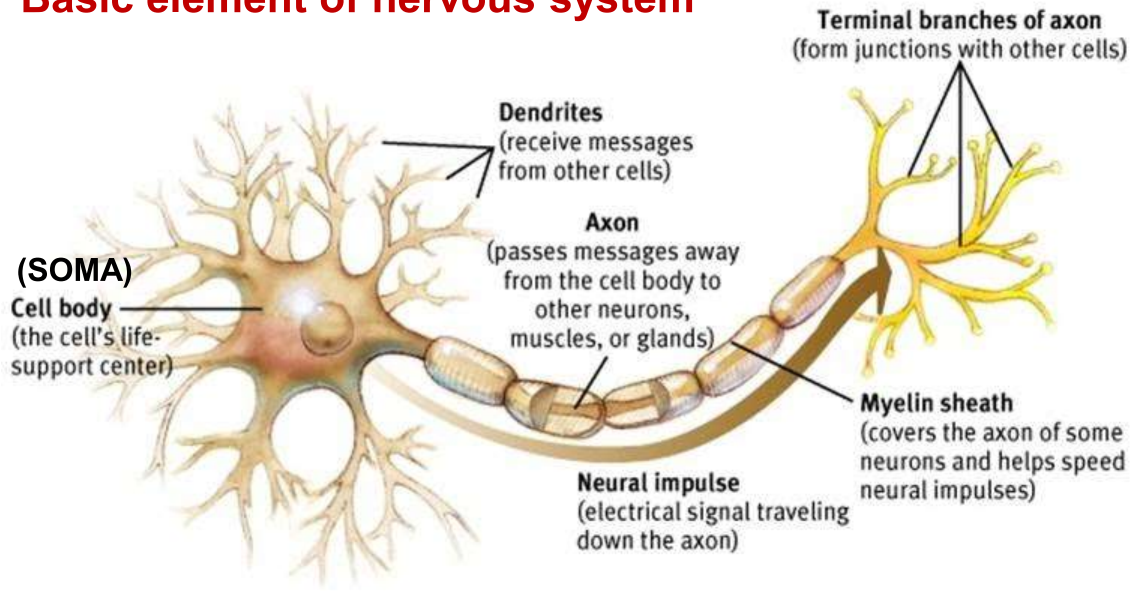
19 July, 2024

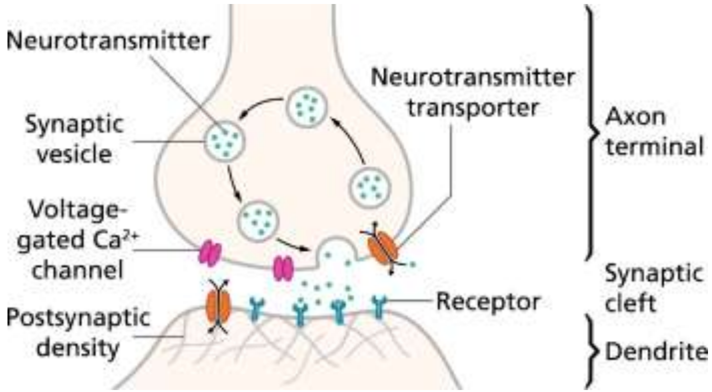
https://www.tbsnews.net/thoughts/marvel-human-brain-unravelling-neural-networks-902791

Biological Neuron

Basic element of nervous system

(SOMA)





Dendrites receive signal from other neuron through **synapses** and pass them on to axon to be transmitted to another neuron.

Synapse is the region at which a neuron communicates with another.

It consists of:

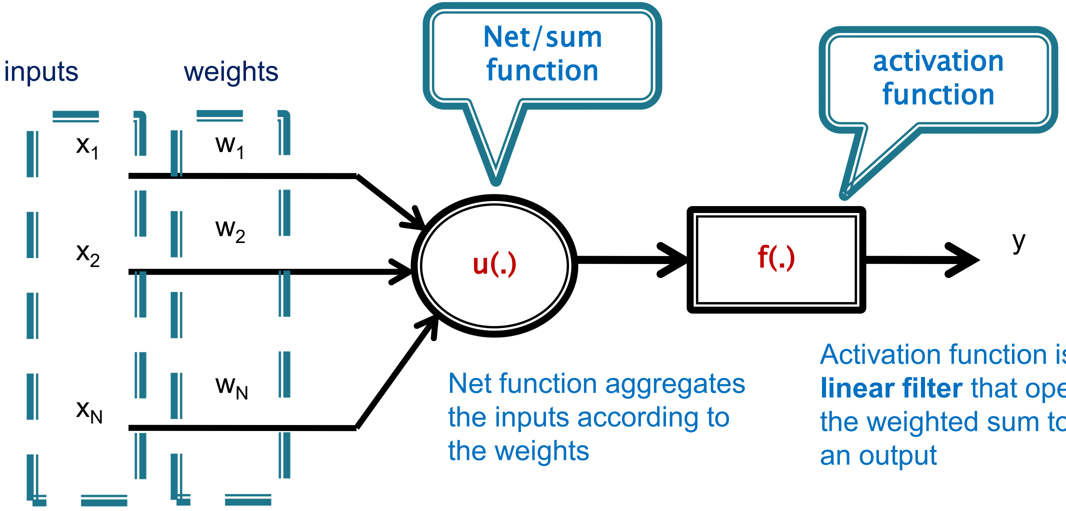
- **dendrites (inputs)**
- **cell body (Soma)**
- **axon (output)**
- **synapse**

A neuron collects signals and **sums** all the '**excitatory and inhibitory**' influences acting on it.

The **excitatory** **dominant the neuron 'fires'** and sends that signal to another neuron through the synapses.

Artificial Neuron

Artificial neuron is a model whose components are analogs of an actual neuron.



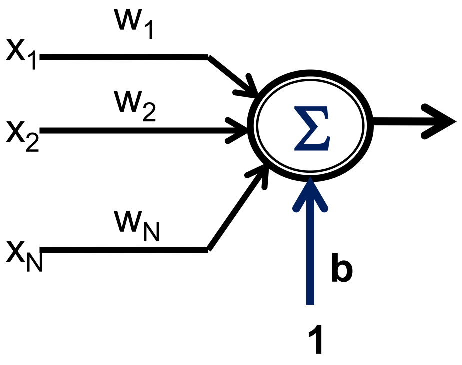
Net function aggregates the inputs according to the weights

Activation function is a **non-linear filter** that operates on the weighted sum to produce an output

Net function and activation function…

Bias or artificial threshold

The **threshold** can also be **replaced** by a constant **input value = 1** with a weight of ***b*** that is equal to the threshold.



$$y = \phi(I)$$

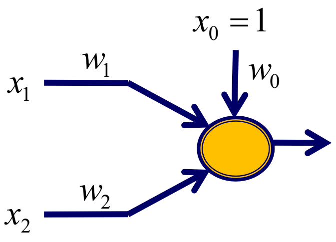
Where, $$I = \sum_{i=1}^{N} w_i x_i + b$$

Perceptron…

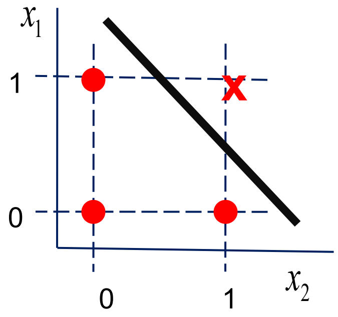
What can a perceptron do?

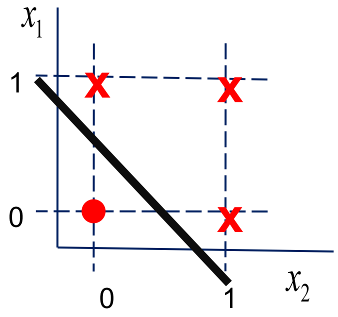
A single perceptron can **solve linear classification** problems.

Example: AND OR types classification by using step activation function



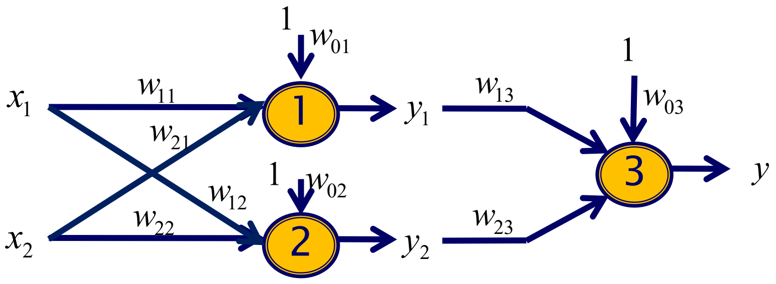
$$y = w_1 x_1 + w_2 x_2 + w_0$$

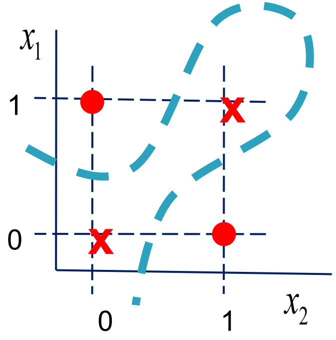




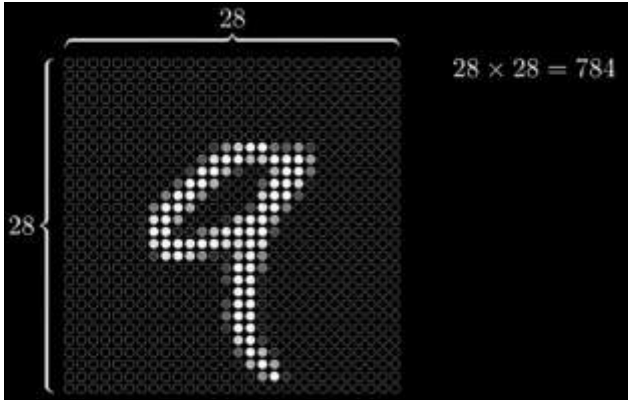
Perceptron…

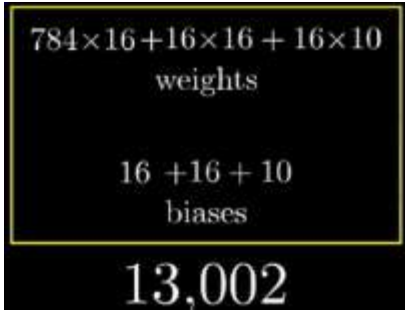
Let's try two nodes for the EXCLUSIVE-OR classification

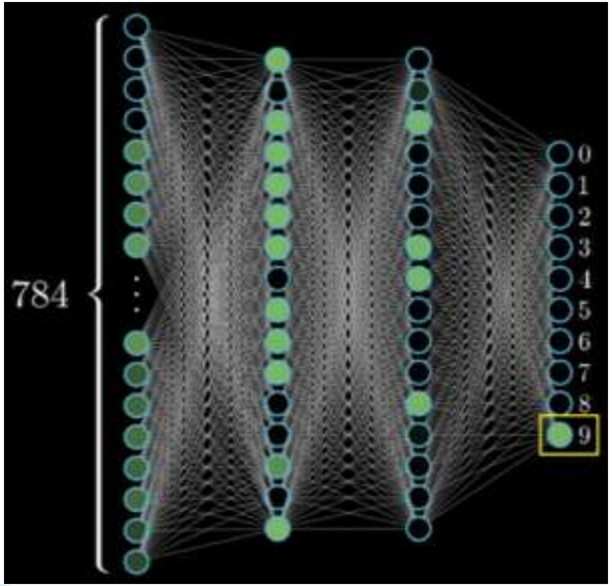




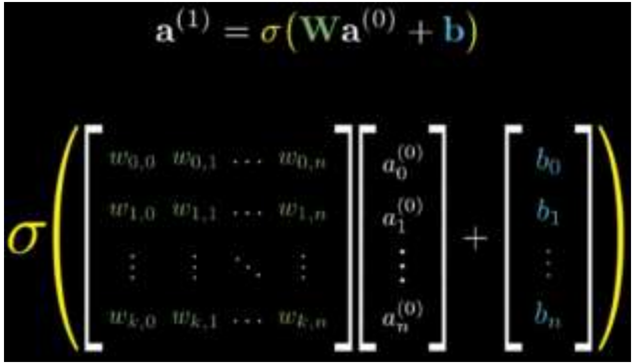
ANN







1st neuron
of 2nd Layer



$$\sigma\left(\begin{bmatrix} w_{0,0} & w_{0,1} & \cdots & w_{0,n} \\ w_{1,0} & w_{1,1} & \cdots & w_{1,n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{k,0} & w_{k,1} & \cdots & w_{k,n} \end{bmatrix} \begin{bmatrix} a_0^{(0)} \\ a_1^{(0)} \\ \vdots \\ a_n^{(0)} \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_n \end{bmatrix}\right)$$

https://www.youtube.com/watch?v=aircAruvnKk&t=990s

GENERATIVE AI

- **Generative AI** is a type of **artificial intelligence** that **focuses** on **creating new content**, such as **images**, **text**, **music**, or **videos**, based on **patterns learned** from existing data.

- It **uses models** that are **trained on large datasets** to understand the **structure**, **patterns**, and **relationships** within the data, **allowing the model** to **generate new instances** that **mimic or extend what it has learned**.

- The **Fourth Industrial Revolution**, marks a **significant shift in industrial operations** by **leveraging advanced technologies** to **automate**, **optimize**, and **enhance production** processes.

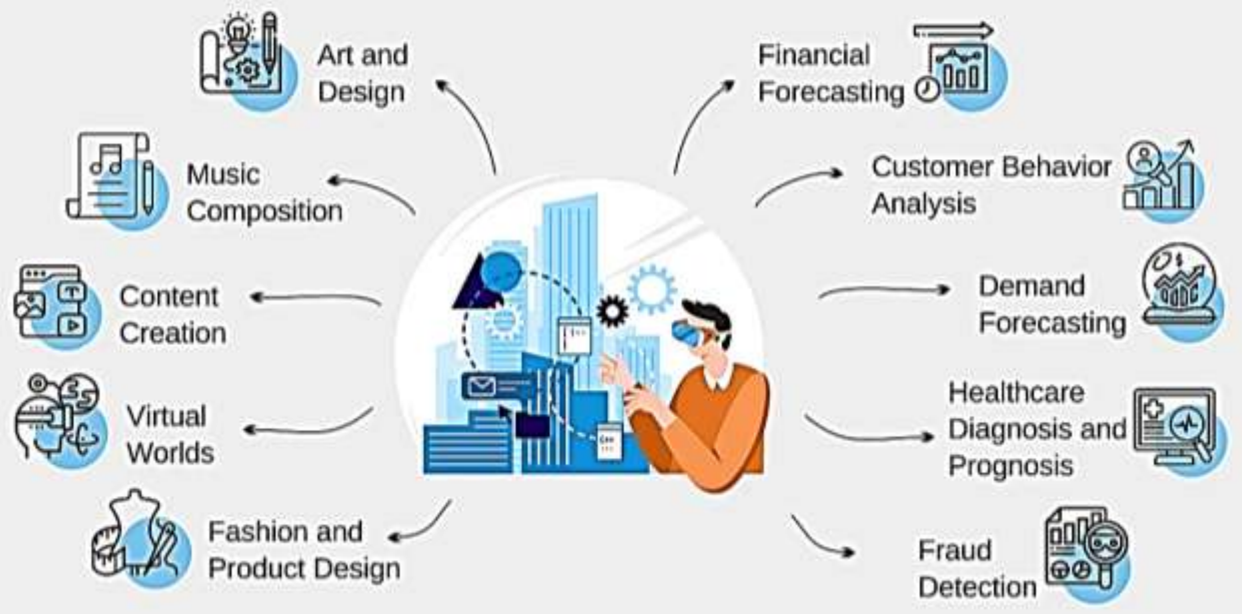
- **IR4 integrates generative AI** as a key component.

Generative AI...

- New image, video, audio or text **could also be generated by other AI** tools or any technologies prevailing. But, the difference between the two is, Generative **AI generated contents are artistic**, creative which could only be performed only by human.
- **Writing poems**, **creative software**, **artistic paintings**, **fashion designs**, **song composition** etc. **cannot be performed by any other techniques or methods**.
- The good examples of Generative AI may be **Dall-E**, **GPT-4** and **Copilot** etc.
- These systems **not only generate creative content** but also **participate** in human like **question-and-answer session**.
- One of the **primary differences** between more **traditional AI and Generative AI** is that the later **can create novel output** that appears to be generated by humans.

Six key Modalities of Generative AI

- Text
- Code
- Audio
- Image
- Video
- 3D/ Specialized



https://www.youtube.com/watch?v=hIIIJt8JERI

- Explainable AI (XAI)

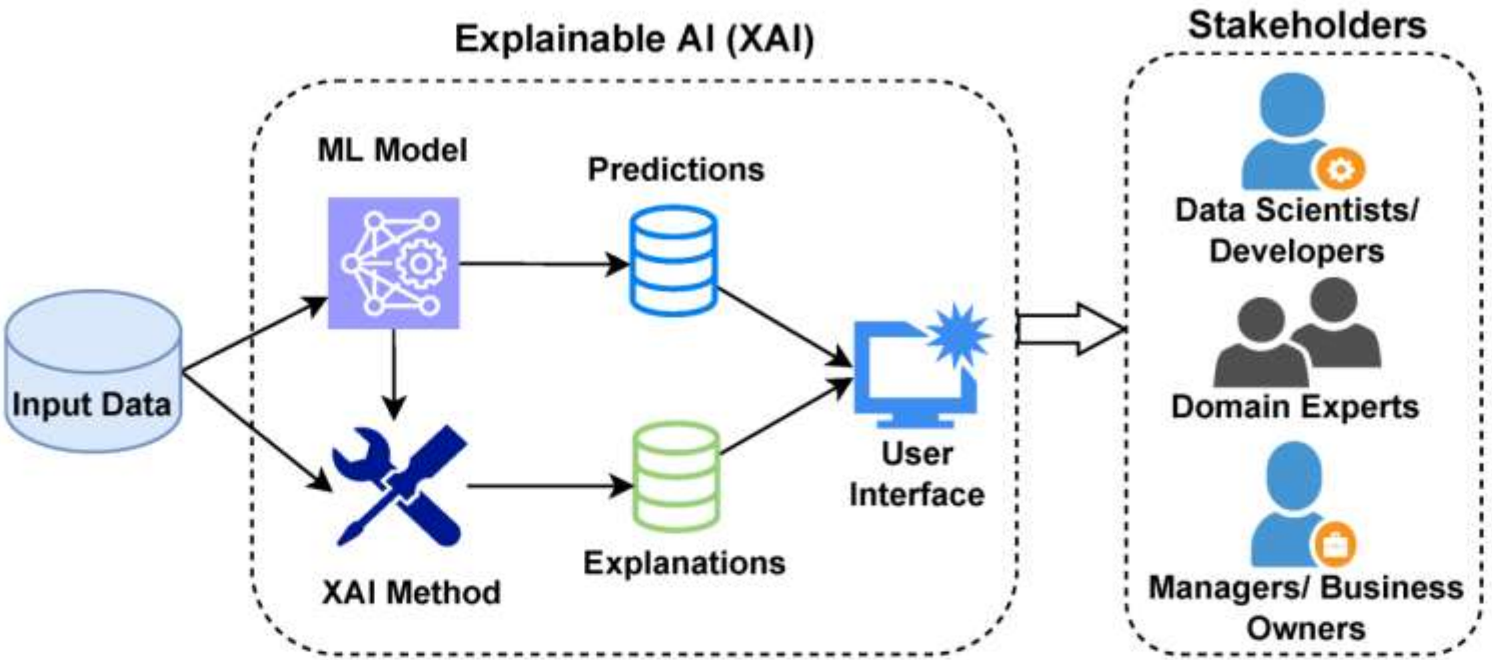
XAI & Industry 4.0

Explainable AI (XAI)

- XAI refers to methods and techniques that make the **output of AI systems understandable** to humans.
- **Aim: Build trust and transparency in AI models.**

AI drives Industry 4.0, but a lack of explainability can lead to doubt and inefficiencies. XAI bridges the gap.

XAI Basic Architecture



Explainable AI (XAI) Overview

- Key Features
  - Offers clarity on how AI makes decisions.
  - Builds confidence by explaining predictions and processes.
  - **Ensures AI outputs** can be evaluated and justified.
  - Enables users to trust and effectively use AI systems.

Explainable AI (XAI) Overview…

- Explainable AI (XAI) refers to systems designed to make AI's decision-making processes **transparent** and **interpretable**

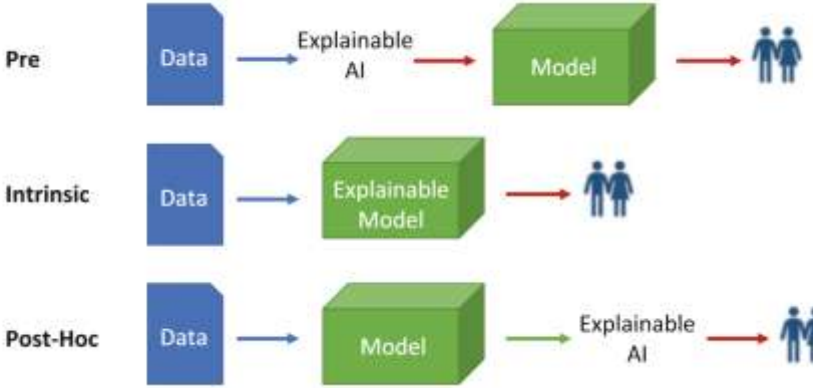
- Unlike traditional "black-box" AI models, XAI ensures that stakeholders understand **how AI systems arrive at conclusions**

- XAI is **crucial for**; building trust and accountability, facilitating regulatory compliance, enhancing adoption in high-stakes domains like healthcare and finance

Traditional ML

XAI – Methods

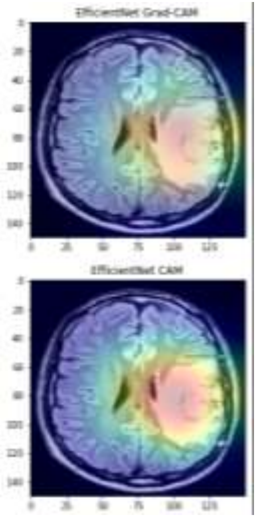
- XAI methods can be categorized based on the **stage of application** relative to the ML Model
  - **Pre-Model:** Exploratory Data Analysis, Feature Engineering
  - **Intrinsic:** decision trees, generalized linear, logistic, and clustering models
  - **Post-Hoc:** visual explanation method, feature importance method, example-based explanation



XAI – Methods…

- **Model-Agnostic Methods**
  - **LIME** (Local Interpretable Model-agnostic Explanations): Provides local explanations for individual predictions
  - **SHAP** (SHapley Additive exPlanations): Explains the contribution of each feature to a model's output

- **Model-Specific Methods**
  - **Decision Trees**: Naturally, interpretable models
  - **Rule-Based Systems**: Explicit rules that can be easily understood by humans

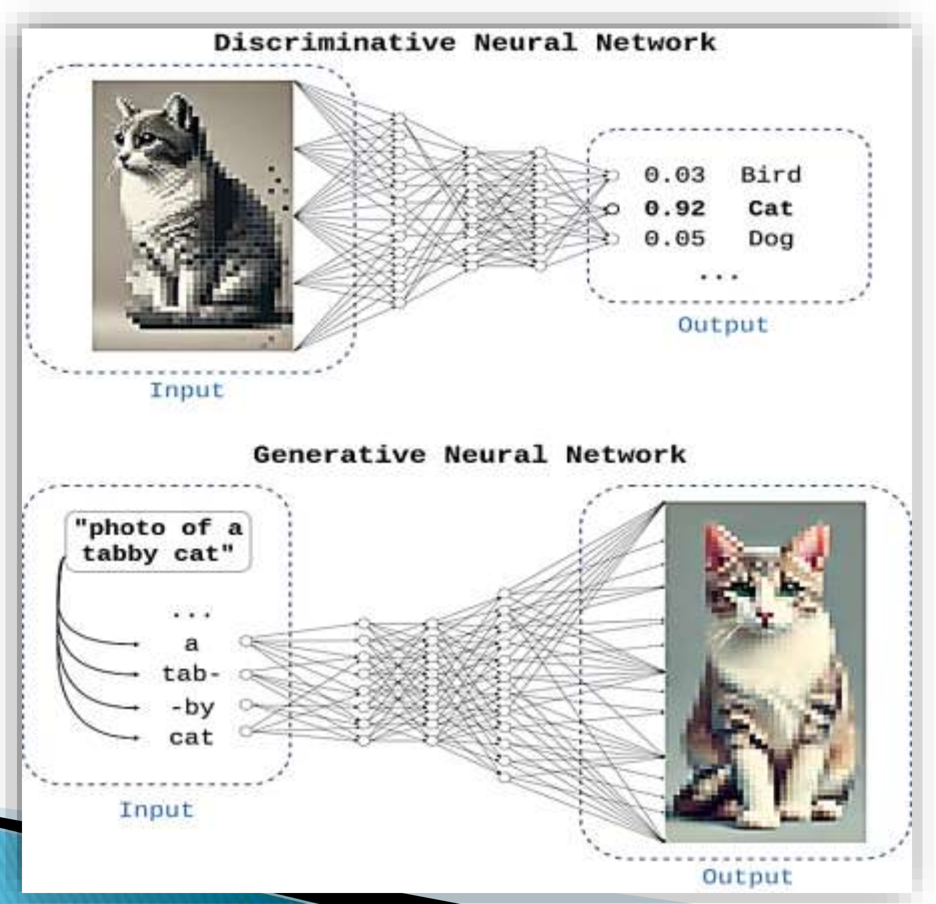


- **Visualization Methods**
  - Visualizations help to translate complex model behaviour into understandable patterns (e.g., saliency maps in deep learning)

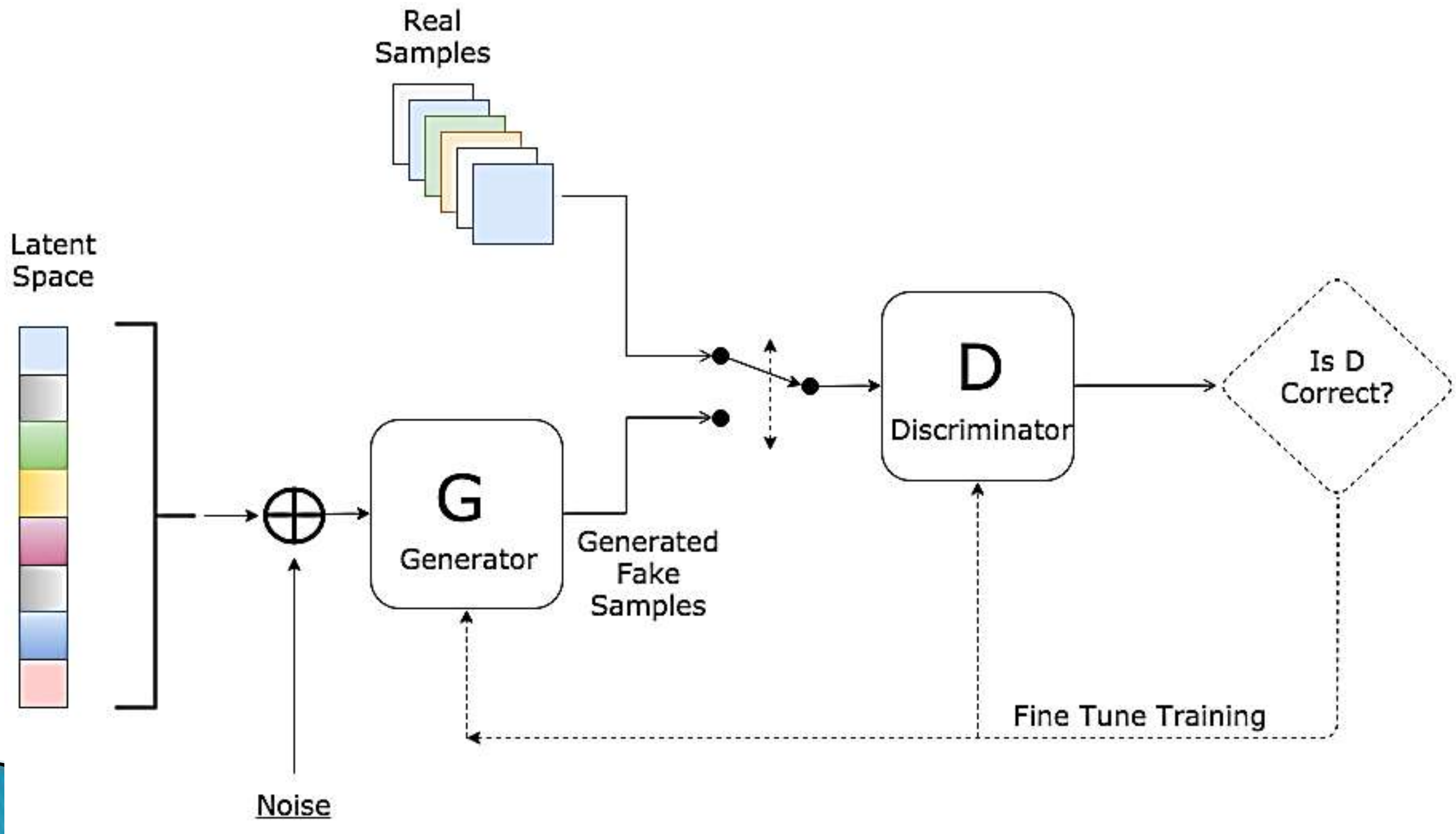
Generative Adversarial Networks

(GAN)

Generative Adversarial Networks



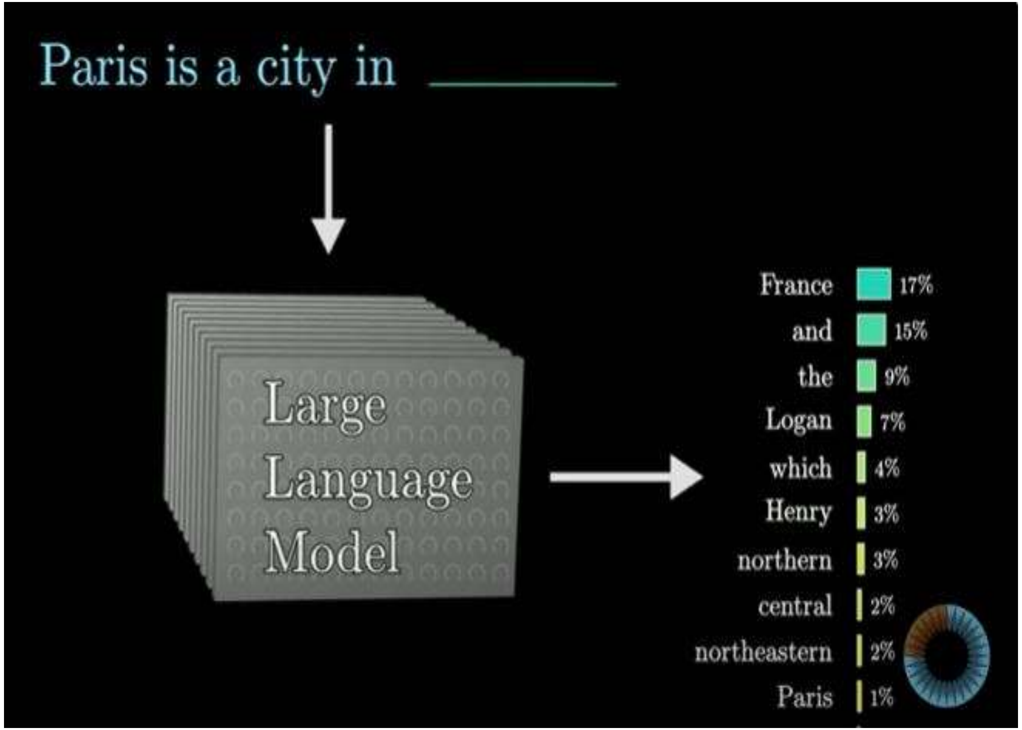
Generative Adversarial Networks…



Large Language models (LLM)

What are LLMs?

- A Large Language Model is a sophisticated mathematical function that *predicts what word comes next for any piece of text*.
- Instead of predicting one word with certainty, though , what it does is assign of a probability to all possible next words. LLMs are capable of generating high-quality text.



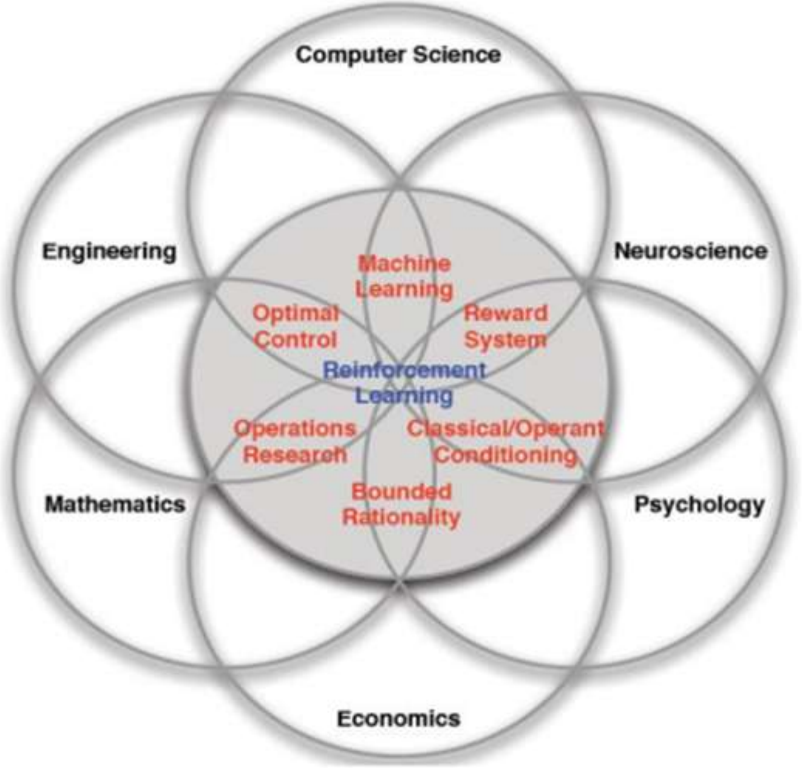
Application of LLMS

1. Chatbot
2. Translation/speech recognition
3. Text Summarization
4. Text Generation/context writing
5. NLP related Tasks
6. Data Analysis
7. Code generation/user experience enhancement
8. Automation and workflow building
9. CSS generation and voice assistants etc.

Reinforcement Learning

Reinforcement Learning

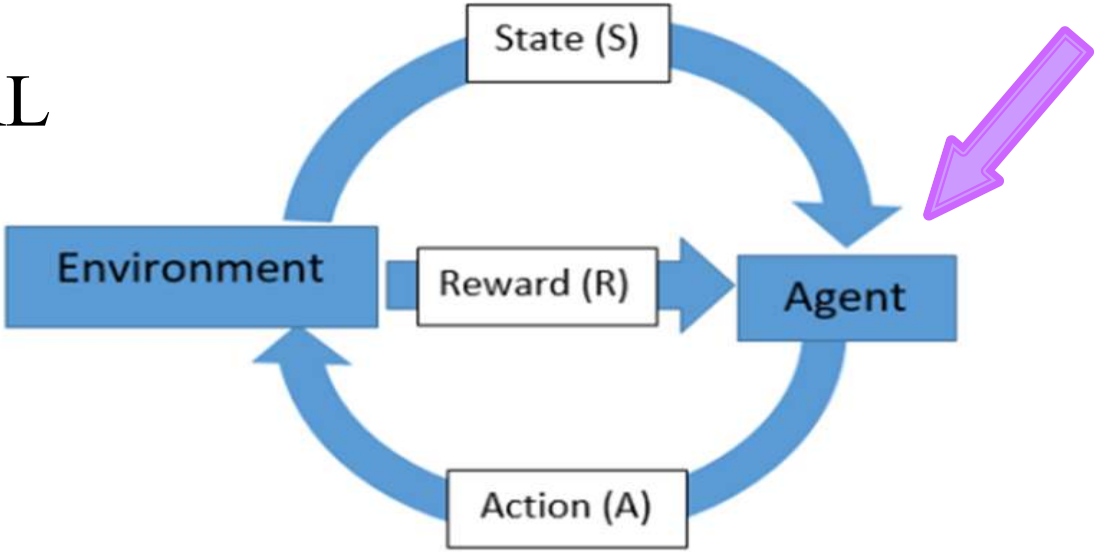
RL, unlike supervised and unsupervised learning ,is learned through interaction with an environment by taking different actions and experiencing many failures and success while trying to maximize the received rewards.



Reinforcement Learning

Main components of RL

1. Policy
2. Reward signal
3. Value function
4. Model



Future of AI

Industry 5.0 and Human-Robot Co-working

https://www.researchgate.net/publication/336816740_Industry_50_and_Human-Robot_Co-working

Future of AI…

High Interaction

Robot Society

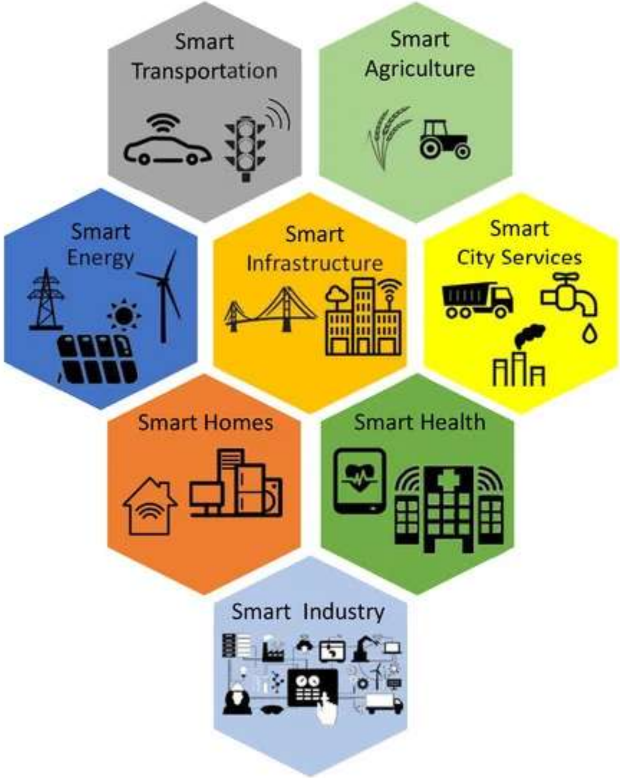
Human Society

Low Control

akhter900@gmail.com

Future of AI…

Smart City

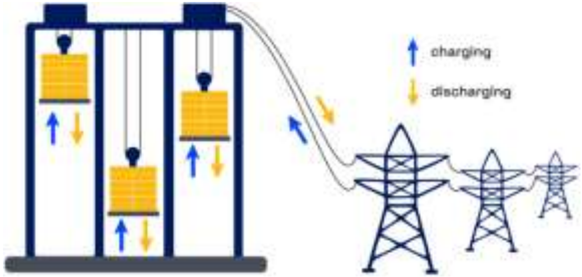


akhter900@gmail.com

Future of AI…

Power Storage

Gravity Battery



Heating Demagnetizes and High voltage shock Re-magnetizes.

Super Capacitor

akhter900@gmail.com

Future of AI…

Quantum ALL

Quantum

Systems

Computer

Computing

AI

Robotics

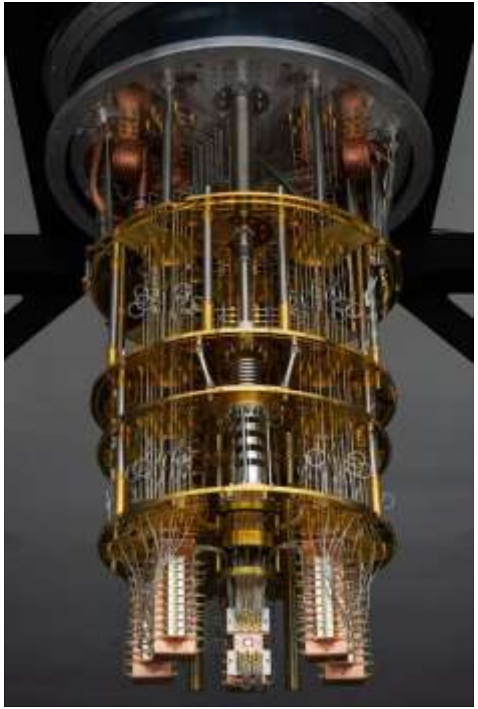
ML

Cryptography

Neuro science

Quantum hardware

… … …



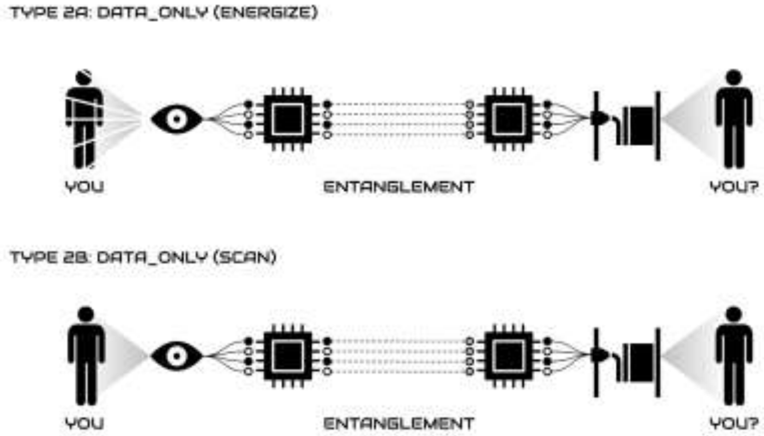
akhter900@gmail.com

Space Intelligence

ELYSIUM

Future of AI…

Teleportation

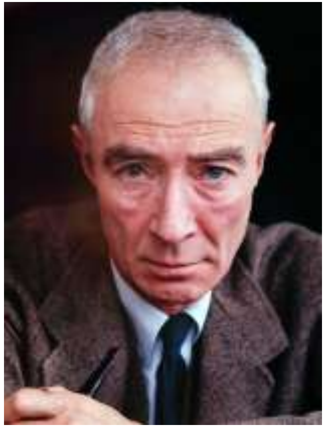


Future of AI…

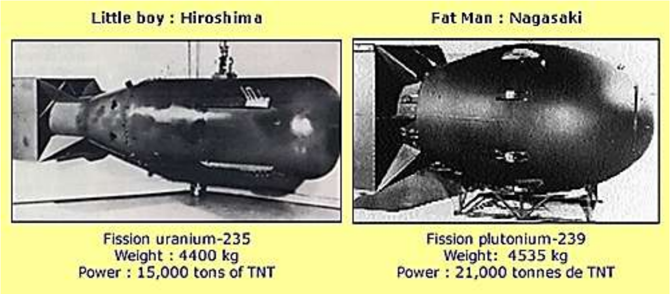
Neuromorphic Computing

Memory Transformation and Systems Consolidation

akhter900@gmail.com




DYNAMITE


Little boy : Hiroshima
Fat Man : Nagasaki
Fission uranium-235
Weight : 4400 kg
Power : 15,000 tons of TNT
Fission plutonium-239
Weight: 4535 kg
Power : 21,000 tonnes de TNT


http://www.

Life is, you verses you, a zero-sum game.

Thank You